

Karthik Bhattaram

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EDUCATION

University of California, Santa Barbara

Santa Barbara

Master of Science in Computer Science

March 2026 to June. 2027

- Courses: Graph Neural Networks, Foundation Models, Distributed Systems, Classic Papers in Security
- Research Area: Network Security, mentored by Christopher Kruegel and Giovanni Vigna

University of California, Santa Barbara

Santa Barbara

Bachelor of Science in Computer Science

Sept. 2022 to Mar. 2026

- Courses: Data Structures and Algorithms, Machine Learning for Networking, Operating Systems
- Honors: Regents Scholar, College of Engineering Honors, USACO Silver Divison

EXPERIENCE

Software Engineer Intern

June 2025 – Sept. 2025

Arista Networks

Santa Clara, California

- Accelerated network switch memory diagnostics by orders of magnitude using custom C++ utilities
- Developed an **interprocess communication** tool for Microsoft and OpenAI datacenters

Machine Learning Research Intern

Sept. 2024 – June 2025

University of California, Santa Barbara

Santa Barbara, California

- Published a paper titled “Leveraging Large Language Models to Contextualize Network Measurements”
- Aggregated FCC data into a SQL database to improve model context using **retrieval-augmented generation**

Software Engineer Intern

July 2024 – Sept. 2024

Cadence Design Systems / OpenEye Scientific

Santa Fe, New Mexico

- Wrote **AWS**-based package to benchmark database compression algorithms that saved terabytes of server space
- Updated **CUDA** C++ software to incorporate new language features to match competing industry solutions

Hardware Architecture Intern

June 2023 – Sept. 2023

Cadence Design Systems

San Jose, California

- Implemented a Python automation tool to generate memory wrappers for use on a hardware emulation platform
- Collected **System Verilog** files to be used as templates for this tool and tested memories it produced

Software Engineer Intern

June 2021 – Aug. 2022

Axiado Corporation

San Jose, California

- Developed a Python package to create **network packet flows** based on IPv6, ARP, ICMP, etc. in simulation
- Created and executed a test plan for a hardware firewall’s parser module and discovered multiple bugs

PROJECTS

Graph Fine Tuning | *Python, PyTorch, Graph Algorithms*

Jan 2026 – March 2026

- Adapted Meta’s Llama 3.1 model to process graph generation tasks using parameter-efficient fine tuning
- Implemented an experimental pipeline to compare zero-shot, Supervised Fine Tuning, and Direct Preference Optimization performance
- Achieved 85% improvement in graph correctness (based on graph metrics) over the zero shot model

TECHNICAL SKILLS

Certificates: Enterprise Security Professional, Python for Data Science and AI

Languages: Python, C/C++, Java, JavaScript, Shell, Verilog

Developer Tools: Git, Visual Studio Code, Vim, Emacs

Libraries: CUDA, Pandas, NumPy, PyTest, Black, Matplotlib